Section Overview

- Throughput, Arithmetic Intensity, and the Roofline Model

Driving Question

RTX 3090		
Peak throughput	35.6	$\mathrm{TFLOP/s}$
DRAM bandwidth	935.8	$\mathrm{GB/s}$

Is there a better performance metric?

We have been measuring our kernels with **latency**, but is there some other metric that better describes "performance"?

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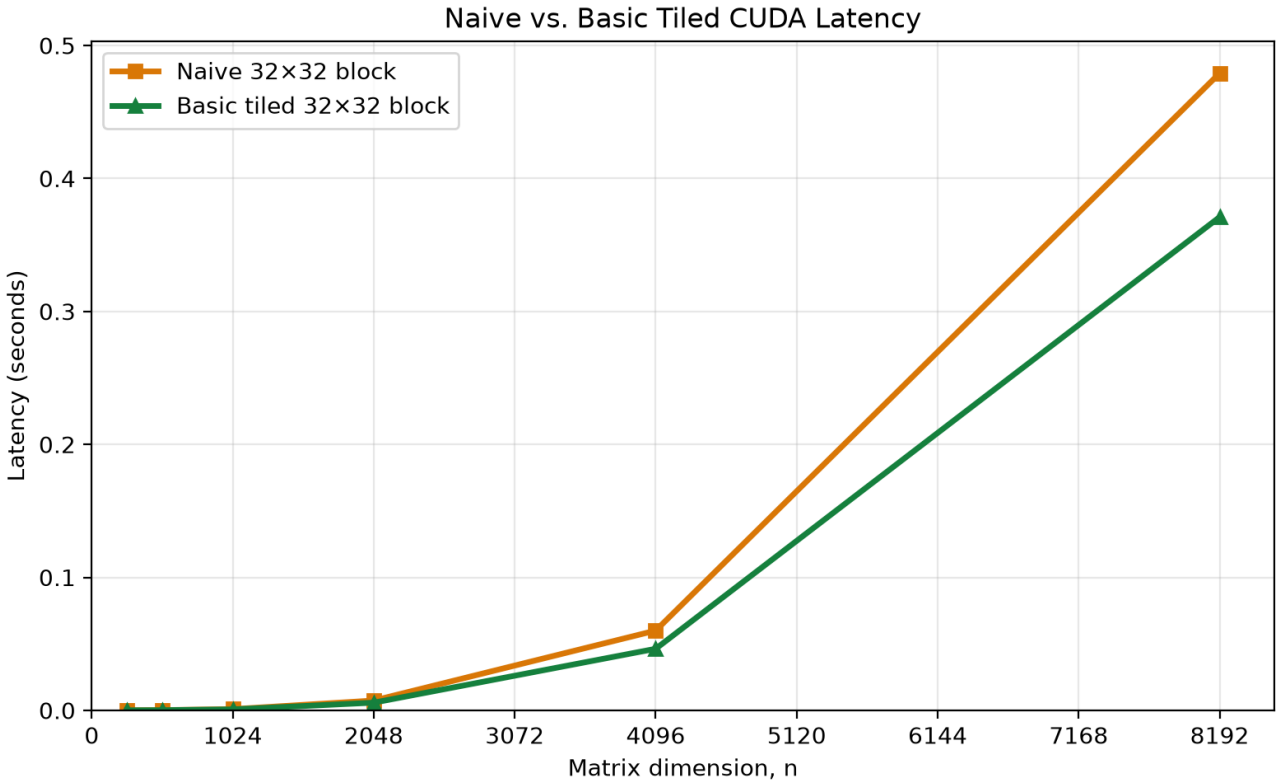
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How fast *can* we be?

Given our hardware constraints, how fast (theoretically) can we do our SGEMM operations?

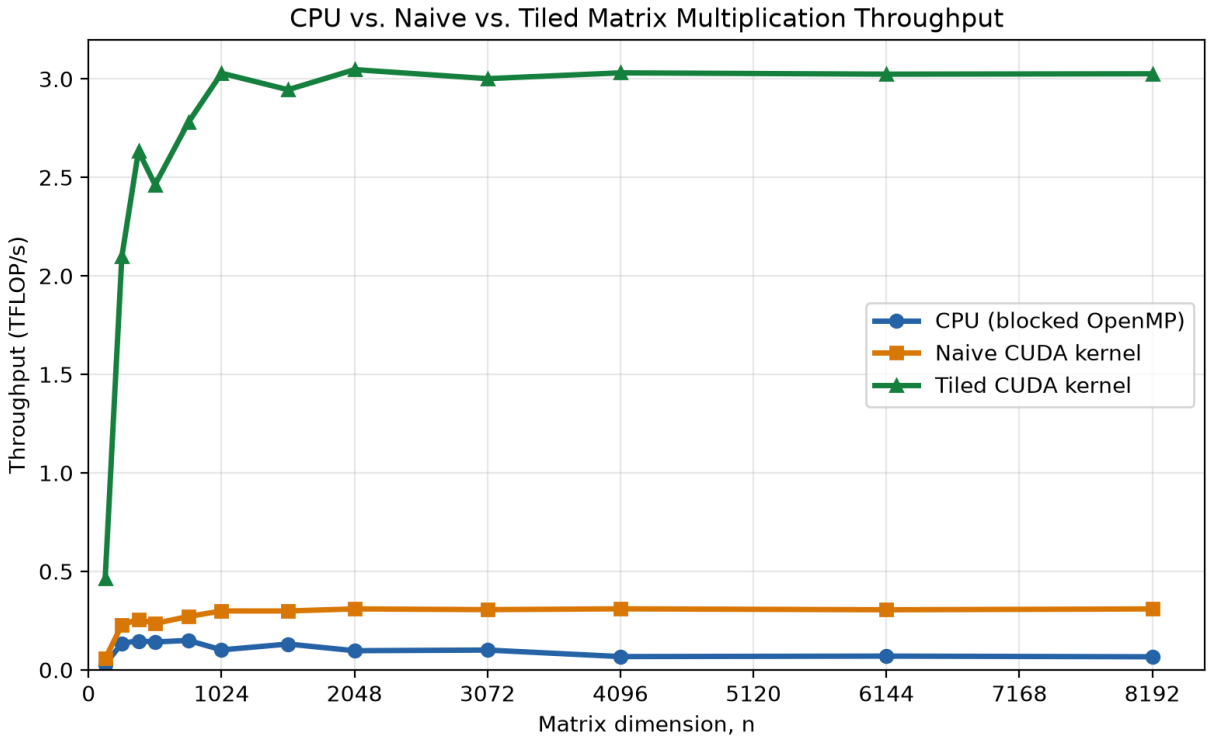
Why latency alone isn't enough

- Latency tells us how long one kernel launch takes, but it depends on the problem size.



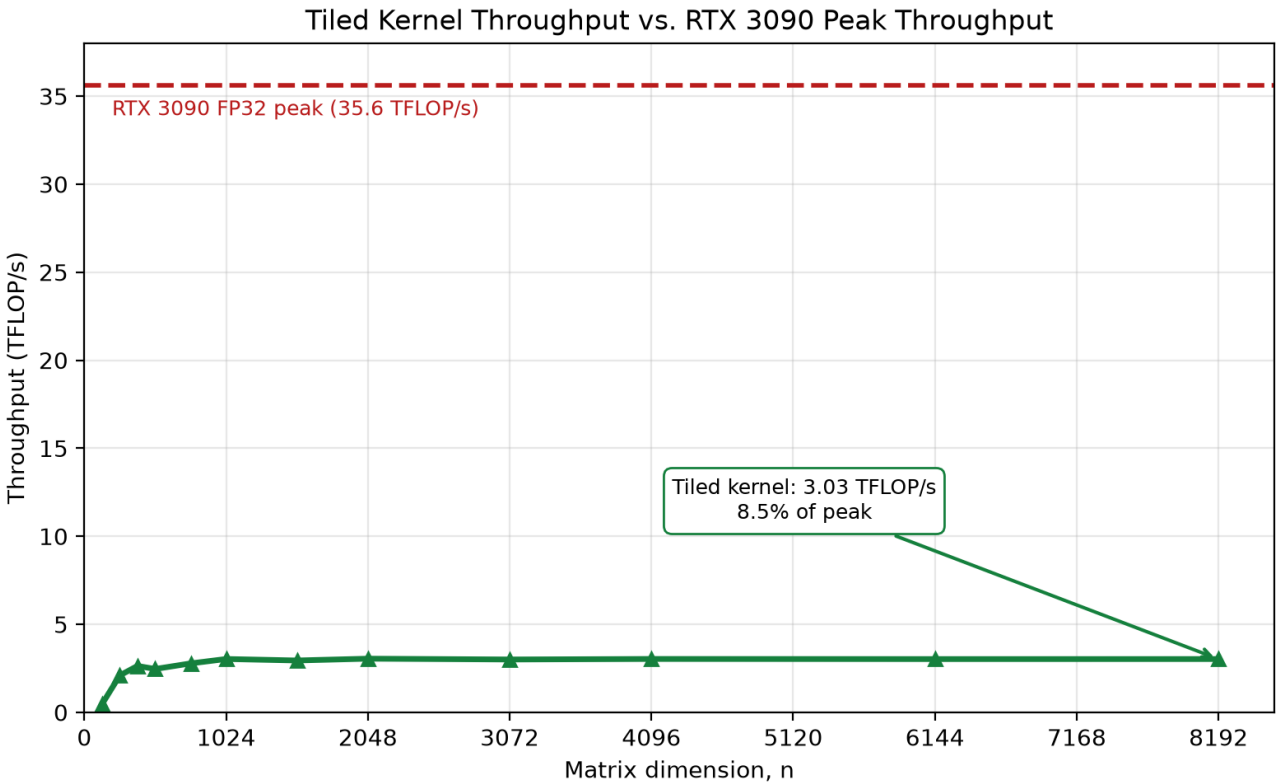
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- Since hardware is characterized by its **peak throughput**, the two values provide a first gauge of how much compute capability our kernel realizes.

Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

Throughput as a performance measure

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Example (Measuring Tiled Kernel)

We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is $2(8192)^3 \approx 1.10$ TFLOPs. Thus, we have an achieved throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\,\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput $35.6\,\text{TFLOP/s}$.

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We observed that we can improve the performance of our SGEMM kernel with tiling, an optimization that improves *data use*. **Can we quantify this data usage?**

Arithmetic Intensity: Characterizing our kernels

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We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

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Example (Naive kernel)

For $n, m, k = 8192$, ignoring hardware L1 caching, we load (from DRAM)

$$n \times m \times 2k \times 4 \text{ bytes} \approx 4398 \text{ GB}.$$

We do 1.10 TFLOPs of work, so we have an AI of

$$\text{AI} = \frac{1.10 \text{ TFLOP}}{4.398 \text{ TB}} \approx 0.25 \text{ FLOP/byte}.$$

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Example (Tiled kernel)

For $n, k, m = 8192$, in the worst case, our tiled kernel loads (from DRAM)

$$\frac{m}{32} \times \frac{n}{32} \times 2\left(\frac{k}{32}\right) \times 32 \times 32 \times 4 \text{ bytes} = 137.4 \text{ GB}$$

With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \text{ TFLOP}}{137.7 \text{ GB}} \approx 8.0 \text{ FLOP/byte}.$$

The Memory Roof

Memory Roof

Given the DRAM bandwidth and a kernel's AI, we say the **memory roof** of the kernel is

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In order for the CUDA Cores to do work, we have to first move the data from DRAM to the SM's. If the kernel only does a small amount of work on the data (low AI), it will likely finish before the next batch of data arrives.

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In order for the CUDA Cores to do work, we have to first move the data from DRAM to the SM's. If the kernel only does a small amount of work on the data (low AI), it will likely finish before the next batch of data arrives. **In this case, the final measured throughput will be constrained by the DRAM bandwidth.** We are **memory bound.**

The Memory Roof

RTX 3090		
Peak throughput	35.6	$\mathrm{TFLOP/s}$
DRAM bandwidth	935.8	$\mathrm{GB/s}$

Example (Tiled Kernel)

Our tiling kernel has AI of 8.0 FLOP/byte, meaning it does 8 floating point operations for every byte of data moved from DRAM.
At 935.8 $\mathrm{GB/s}$ of bandwidth, our **memory roof** is

$$935.8\,\mathrm{GB/s} \times 8.0\ \mathrm{FLOP/byte} \approx 7.49\,\mathrm{TFLOP/s}.$$

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$$935.8\,\mathrm{GB/s} \times 8.0\ \mathrm{FLOP/byte} \approx 7.49\,\mathrm{TFLOP/s}.$$

This is much lower than the peak throughput of $35.6\,\mathrm{TFLOP/s}$. **No matter how we optimize this kernel, without increasing AI, we will never achieve greater than** $7.49\,\mathrm{TFLOP/s}$.

Hypothetical Kernel

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Example

Imagine we have some kernel that has double the AI at 16.0 FLOP/s.

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$$935.8\,\mathrm{GB/s} \times 16.0\ \mathrm{FLOP/byte} \approx 15.0\,\mathrm{TFLOP/s}.$$

Since $15.0\,\mathrm{TFLOP/s} < 35.6\,\mathrm{TFLOP/s}$, this kernel will still be memory bound.

Hypothetical Kernels

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AI (FLOP/byte)	Memory Roof (TFLOP/s)
8	7.49
16	14.97
32	29.95
36	33.69
38	35.56
40	37.43

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38	35.56
40	37.43

Wait! Our peak throughput is 35.6 $\mathrm{TFLOP/s}$! At AI ≥ 38, we are **compute bound**. Our data can finally keep up with the CUDA Cores!

Ridge Point

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DRAM bandwidth	935.8	$\mathrm{GB/s}$

AI $= 38$ FLOP/byte is known as the **ridge point** *RP*, and can be found by solving:

$$\text{Memory Roof} = \text{Peak throughput}$$

$$\text{DRAM Bandwidth} \times \text{RP} = \text{Peak throughput}$$

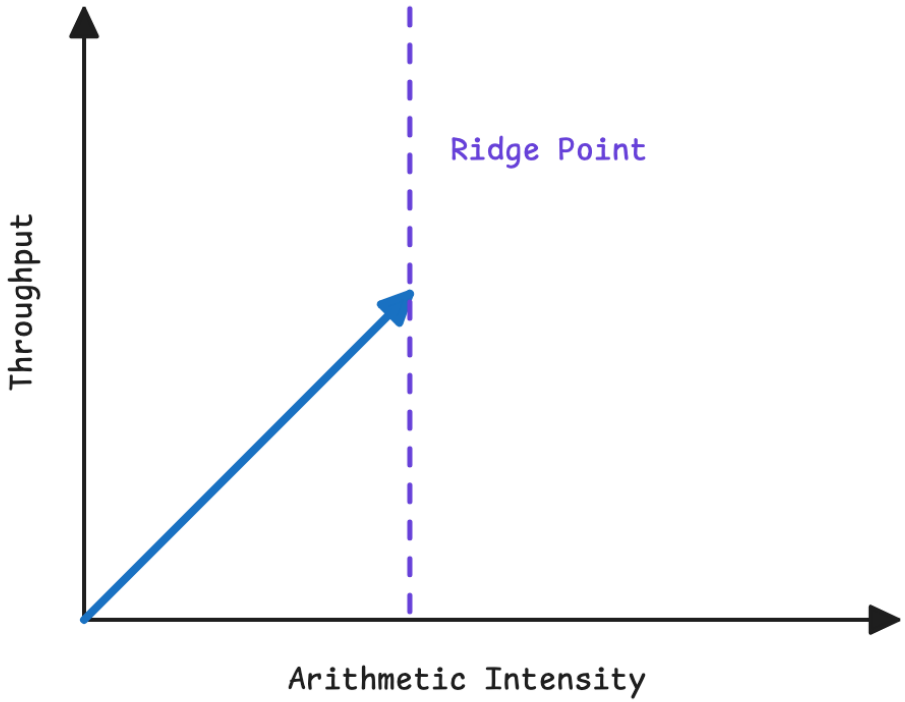
$$\text{RP} = \frac{\text{Peak throughput}}{\text{DRAM Bandwidth}}$$

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Roofline Model

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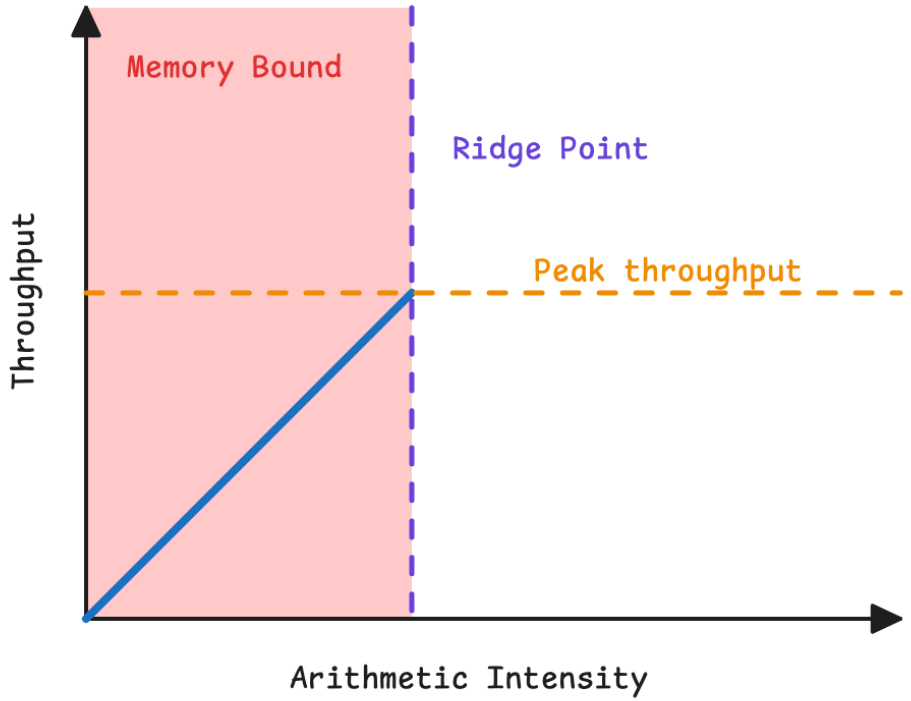
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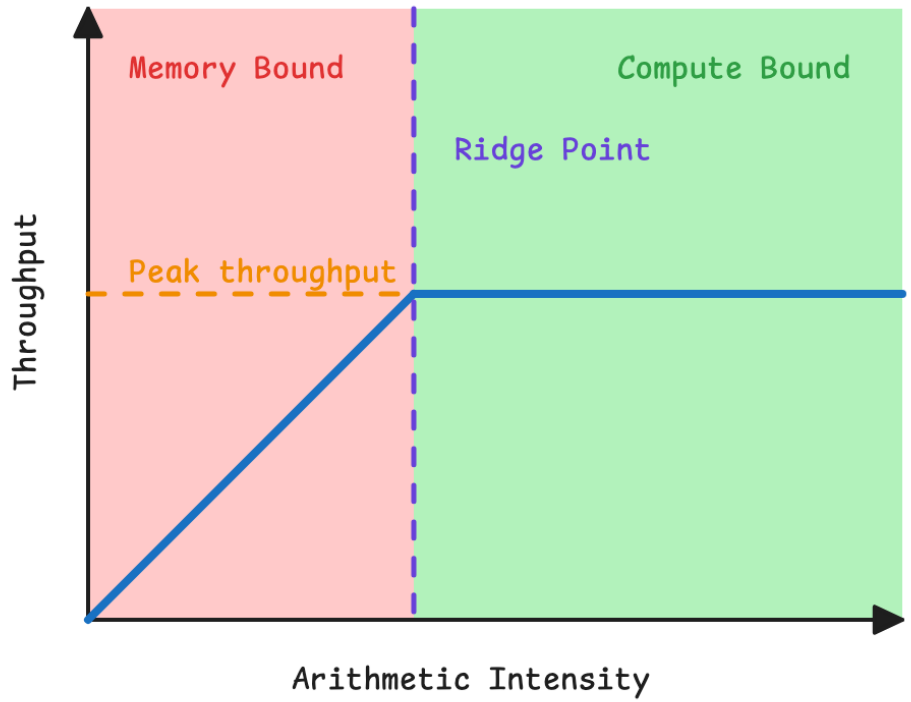
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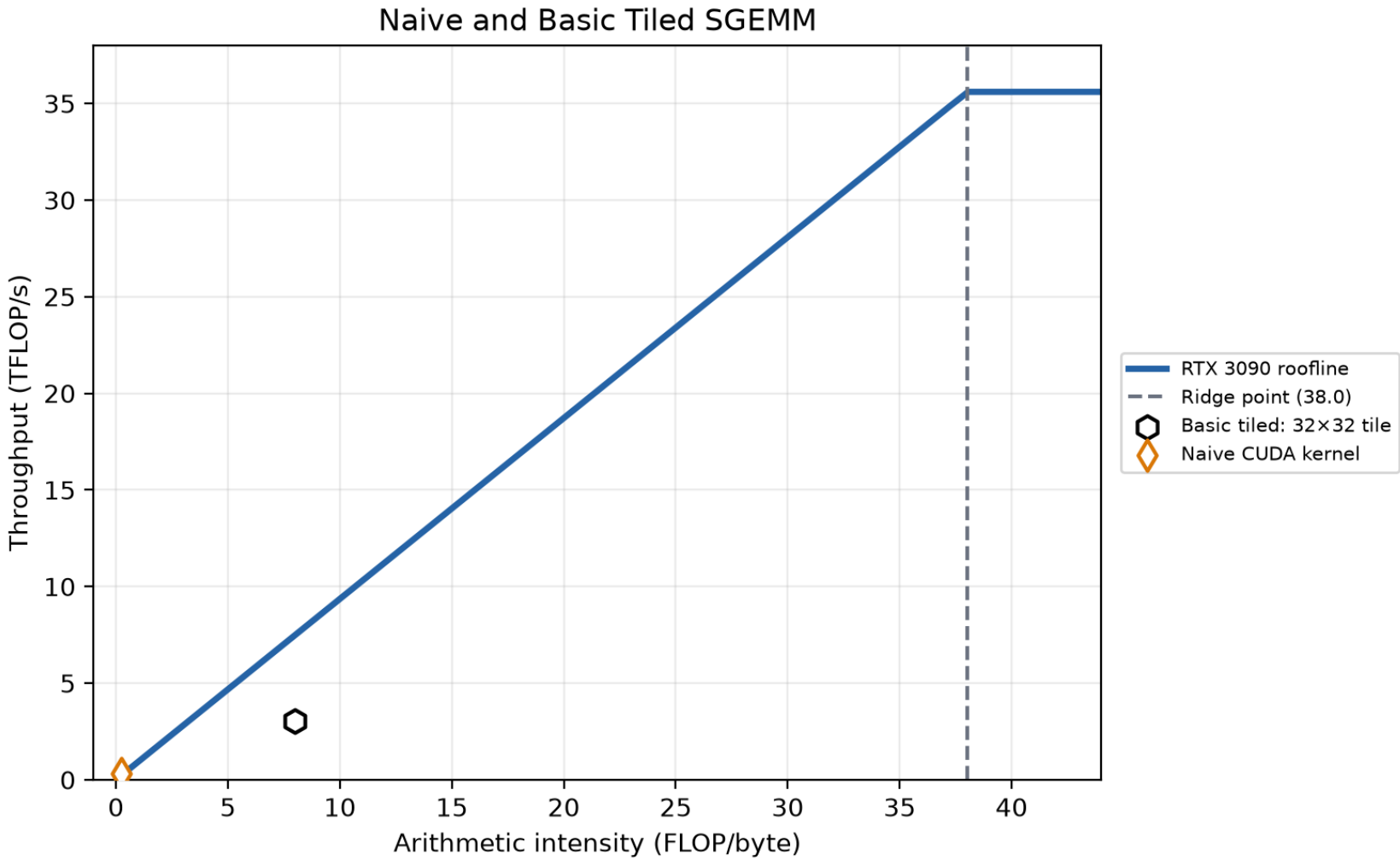
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Roofline Model of our RTX 3090



Since we are memory bound, increasing AI should increase performance.